



- 7 A particle P is projected from a point O on a horizontal plane and moves freely under gravity. The initial velocity of P is 25 ms^{-1} at an angle θ above the horizontal, where $\tan \theta = \frac{4}{3}$. At point A , the direction of motion of P makes an angle of 45° with the downward vertical through A .

(a) By differentiating the equation of the trajectory or otherwise, find the coordinates of A . [5]

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At point A , the particle strikes a fixed smooth barrier, rebounds, and lands on the horizontal plane. The barrier is inclined at an angle of 45° to the horizontal.

- (b) Find the speed of P immediately before it collides with the barrier. [3]

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- (c) Given that the coefficient of restitution between the barrier and the particle is $\frac{1}{9}$, find the horizontal distance travelled by P after it strikes the barrier. [4]

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